Ozone Concentration Prediction

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Introduction

Ozone is an air pollutant which can cause harmful respiratory and other health effects on humans [1]. Predicting ozone concentrations in different conditions is useful for ozone control and public health management. Ozone concentrations depend on both weather conditions (wind, temperature, etc.) and emissions; especially VOC (volatile organic carbon) and NO_x (NO_2 + NO). These interactions can be empirically modeled using machine learning models trained by observation data to predict ozone levels over time [2]. Furthermore, multiple models can be utilized to predict ozone concentration between weather observation stations or even to forecast ozone levels under different weather and emission conditions.

Problem Definition

We will focus on the South Coast Air Basin (SoCAB) region in California from 1980 to 2020. Since myriad variables potentially affect ozone concentration, we will identify which variables are more important for predicting. Also, relative influence of these identified variables on ozone levels will be studied using different machine learning models. Based on empirical model established by machine learning methods, spatial distribution of ozone in SoCAB region and ozone isopleth (ozone concentration under different VOC and NO_x emission levels) can also be studied.

Data Collection

Our project focus on SoCAB region in California. To train ozone empirical model, we include several features which have potential relationships with ozone concentration.

Emissions

Emissions of VOC and NO_x for each year are estimated by previous research [2] (Figure 1). Obviously, emissions of VOC and NO_x decrease in these years while the decrease rate is lower in recent years especially for VOC.

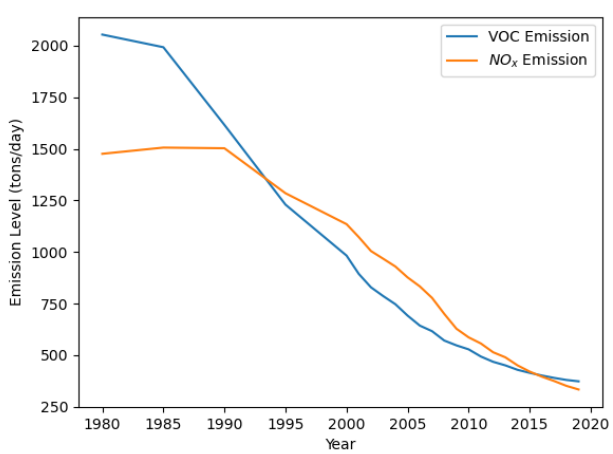

Figure 1. Emissions of VOC and NO_x in SOCAB

Ozone

We use hourly ozone concentration data from EPA website. Based on Social boundaries, 68 monitors are included in the project. The spatial distribution of monitors is shown below. Since previous research shows that the yearly fourth high 8-hour maximum ozone (ozone design value) has relative significant effect to human health, we calculate ozone design value (select maximum value of all 8-hour consecutive ozone for each day and then find the fourth high for whole year) for each year each site and use them as labels in our machine learning model. Spatial averaged ozone design values are shown below. Ozone concentration is decreasing during these years while it fluctuates in recent years.

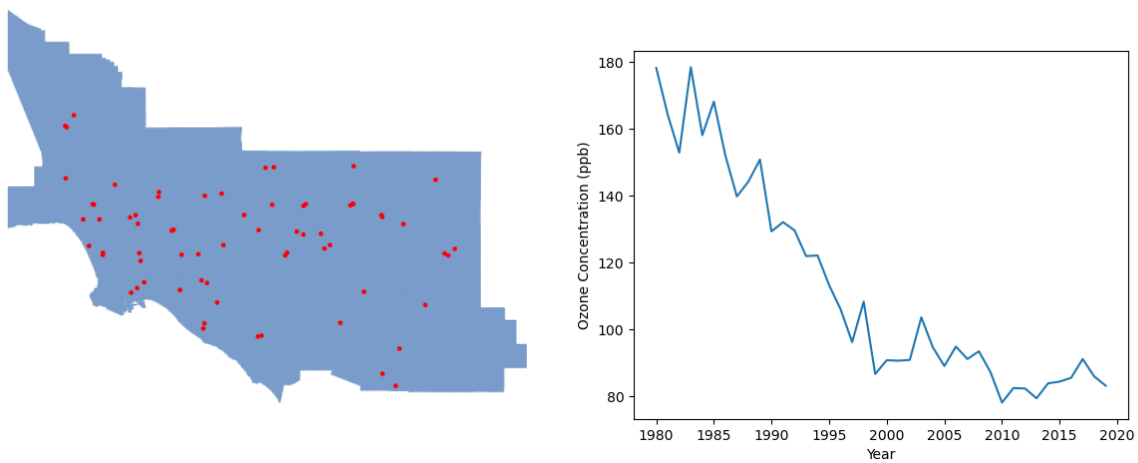

Figure 2. Distribution of selected monitors in SoCAB (left figure). Spatial averaged ozone concentration in SoCAB from 1980 to 2019 (right figure).

Meteorology and Topology Features

Since different locations and meteorology conditions also affect ozone formation and deposition, we include meteorology and topology features into our model. For topology data, we extract elevation for each site from LANDFIRE website which provides high resolution topology data. Also, we include latitude and longitude of each sites which can be used to train the model and analyze spatial distribution of ozone in model application. Julian day and hour for ozone design value are selected to describe sun flux which may affect photochemistry of ozone.

For the meteorological data, we mainly focus on four features, which are Winds, Temperature, Barometric Pressure and RH. We download all the data from EPA website. To make our features more convincing and scientific, we decide to use the hourly data and calculate the average of maximum 8 hours for all of the meteorological features. However, there still exists missing data since the EPA uses different site ids to monitor each of the features. To make our data consistent, we use the site which has the minimum distance to the ozone monitors and capture the data from the closest meteorology site for the missing data and directly replace it.

Methods

The ozone concentration and weather data are provided by the EPA. Since there are several missing features for the data, we dropped the data which has more than two missing features and then k-Nearest Neighbors imputation method is conducted to estimate missing features. A weather simulation model based upon satellite data is a potential data source when we apply our model to study ozone spatial distribution since the simulation model could provide fields of meteorology parameters. Emissions of VOC and NO_x are estimated by previous work [2]. Also,

some geographic information such as elevation, latitude, longitude, and time will be included since these affects solar flux, which is an important driving force for chemical reactions in the atmosphere. In this project, we will provide two routines to train the empirical models. First, we will use principal component analysis to reduce the dimensions of our data and then use polynomial regression on the reduced data to predict ozone concentrations. Backward elimination method and LASSO regression are conducted to reduce polynomial terms for the regression analysis. The other method we will use is a self-organizing map to reduce dimensions and then an artificial neural network to predict ozone levels [3]. We can potentially compare these two model training routines by using all features directly. Random forest is another potential algorithm since it has better performance for discrete relationships than the supervised algorithms we listed above [4]. To avoid overfitting and estimate the uncertainty of our model, data withholding will be conducted.

Results and Discussion

We extract 1171 data points from data sources while 206 of them are missing at least two features. These data are most concentrated in early year due to limited monitors in California. We drop these data and conduct 3 nearest neighborhood imputation method to estimate the missing values for the left data.

The covariance matrix between each feature is shown below. NO_x and VOC emissions have the highest correlation values with ozone among selected features. It is understandable since VOC and NO_x emissions are two dominant chemical species which affect ozone concentration [6].

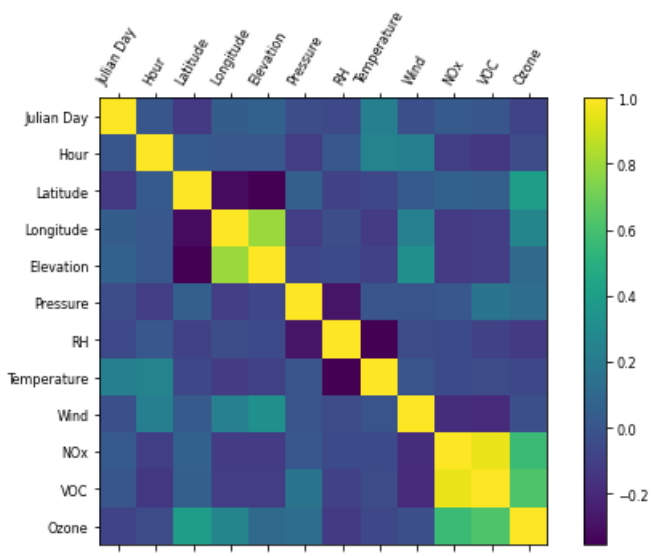

Figure 3. Correlations between features.

Principal Components Analysis and Polynomial Regression

Principal Components Analysis

Firstly, we use principal components analysis to reduce the dimension of our data. The ratio of explained variance for each component is shown in figure 4. The highest three features are explained 99.19% variance. So, we extract these three components and to find the dominant features in the principal component.

The principle components are the linear combination of original features. From figure 5, NO_x, VOC, pressure, elevation have the highest absolute coefficients. It means that these features can reproduce principal components without too much loss. So, we select these features for training polynomial regression model.

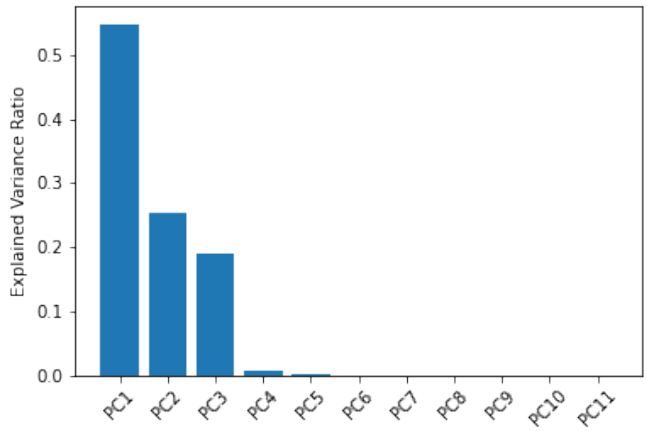

Figure 4. Ratio of explained variance for different components.

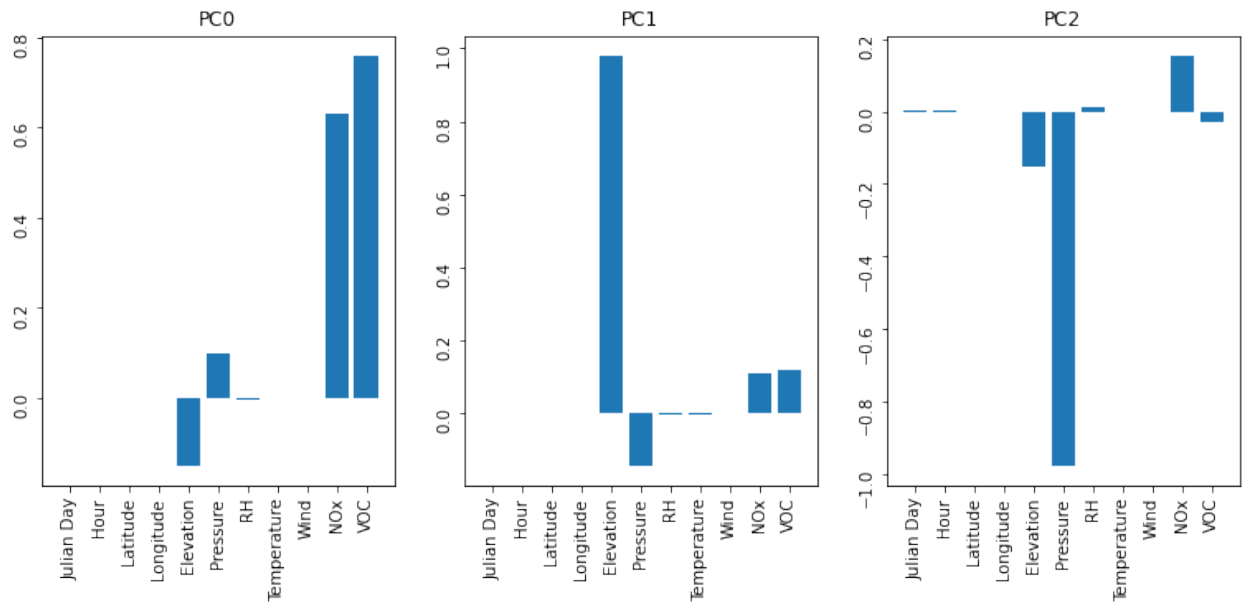

Figure 5. Ratio of explained variance for different components.

Polynomial Regression

Second order polynomial regression is conducted to predict ozone under selected feature in PCA step. There are 14 terms which order is lower than 2. Backward elimination method is conducted to eliminate these terms. Nine of them are remained after elimination and polynomial regression is applied by using remained terms and labels (ozone concentration). The terms we remain are:

Table 1: Remained terms for polynomial regression (E: elevation, P: pressure, N: NO_x, V: VOC)

Term 1	Term 2	Term 3	Term 4	Term 5	Term 6	Term 7	Term 8	Term 9
E	P	V	E^2	EP	P^2	PV	N^2	NV

Since backward stepwise method uses p-value as metric for each term, some terms with low p-value could have high importance when they combined with other terms. LASSO regression is a good method to avoid this limitation. We conduct LASSO regression for feature selection with hyperparameter α=0.1 and the result is as same as the backward stepwise method (Table 1).

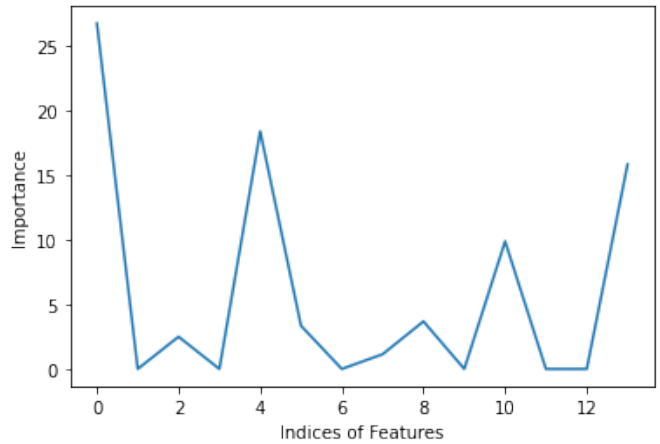

Figure 6. Importance of each term estimated by LASSO regression.

Regression performance is shown in figure 5. The trend and peaks are captured in regression model while the bias is relatively high in early year (high ozone concentration) and low in recent year (low ozone concentration) which is favorable since stakeholders are more interested to predict ozone concentration for the future which is expected as low ozone concentration.

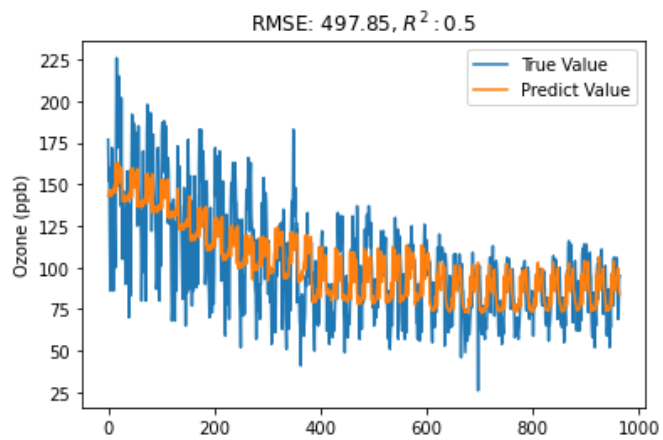

Figure 7. Polynomial regression result with selected terms.

To estimate uncertainty of our model and avoid overfitting problem, we use 10-fold cross validation method. R^2 is 0.49 and RMSE is 503.60 which is close to the polynomial regression result trained by all dataset. It means that our model performs well even when some data is withheld.

Self-Organizing Map and Neural Network

Feature Extraction before NN

Before we start Neural network analysis, we apply decision tree algorithm and a self-organizing map to extract important features.

Decision tree will automatically choose a feature which makes a great contribution to the model based on information entropy, which means if a feature is selected by more subtrees, it is more important.

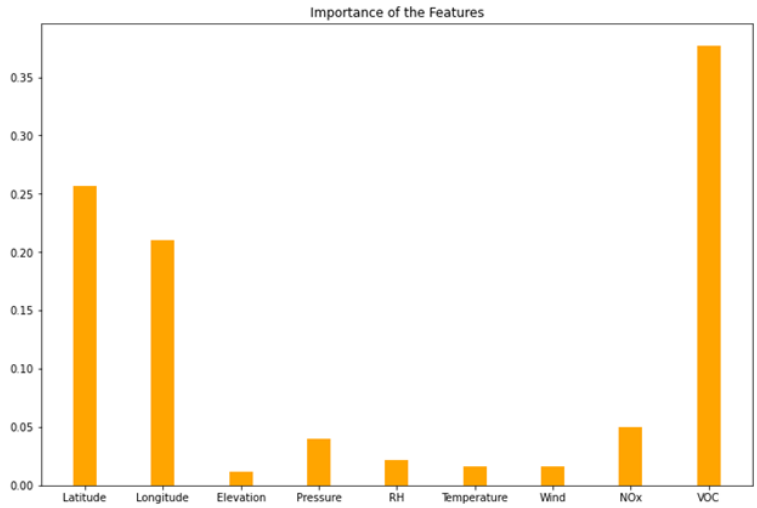

Figure 8. Feature importance from decision tree algorithm

Since we collect 9 features as our dataset, it is likely that several variables could have inter-relationship. For instance, latitude and longitude will affect the emission of VOC and NOx strongly. According to Hoeffding's inequality, too many features will lead to overfitting due to the increase of hypothesis set. In this part, we use self-organizing map algorithm to delete the features which are not independent in the real world.

Normally, SOM is used in reducing high-dimension data into 2-dimensional plots which can also store the topological structure of the original dataset. After several tests and attempts, we decide to use size = $5\sqrt{N}$, learning rate = 0.1, sigma = 1, neighborhood function = 'gaussian' as the parameter. We plot the weight vector for each neuron below. If two plots are highly similar, we can consider that they have strong inter-connection, which means we should pick only one of them as an input for our neural network. After comparing the weight factors of different features, we will use 'Temperature', 'VOC', 'NOx' and 'RH' in our neural network training.

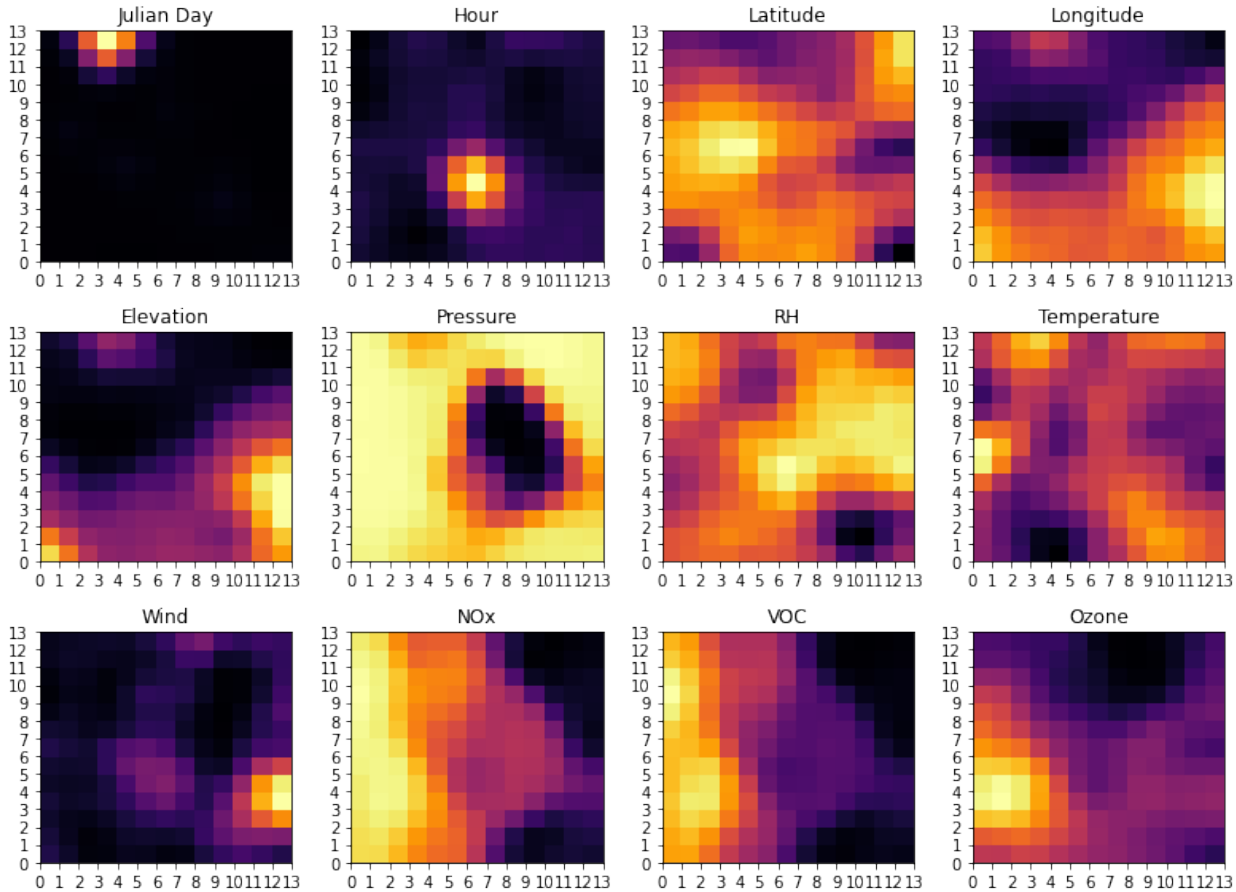

Figure 9. Weight vector plot for each neuron

Neural Network

Based on the results of SOM, we use the library sklearn.neural_network on the selected features 'Temperature', 'VOC', 'NOx' and 'RH'. Considering different units for each feature, we use the function StandardScaler() to scale the dataset. Next we build a neural network model with 3 hidden layers with 256 neurons in each layer, activation function "relu" and max_iter=4000. We train the model by inputting all data, the R^2 is 0.30 between labels and predicted labels and RMSE is 386.57 (figure 10). We also use MLP regression score to evaluate our neural network which is 0.61. To estimate uncertainty and avoid overfitting, we implement 10-Fold cross validation to analyze the performance of the built network model. The cross-validation scores are as follows: [0.64304825, 0.65980842, 0.47703865, 0.47882531, 0.257692, 0.63802222, 0.4268952, 0.55780633, 0.4537101, 0.50968802]. Averaged R^2 and RMSE are also calculated which are 0.17 and 470.23 separately.

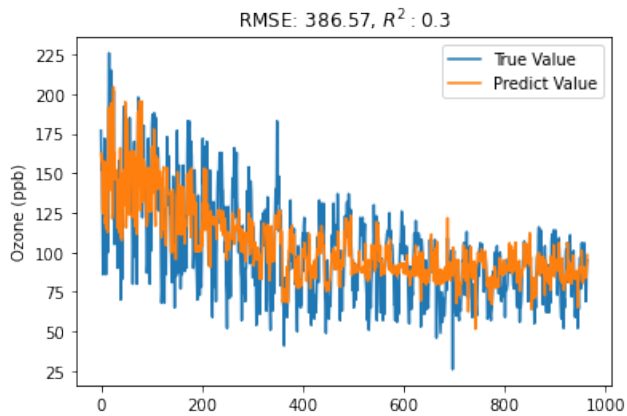

Figure 10. Actual Ozone values and predicted Ozone values.

Random Forest

The random forest method is an ensemble supervised learning technique that runs many decision trees in parallel and averages the predictions to create a more accurate final regression. We used this algorithm to attempt to predict ozone concentrations based upon all other parameters from the data (except for station number and date, which were removed before regression).
The data was split into 10 random sections (tenfold cross validation) to prevent overfitting and then the hyper-parameters of tree number and maximum tree depth were iterated in nested for loops in order to use the elbow method to find the best fit. See graphic below for the elbow method selection of 5 estimators and a max depth of 5.

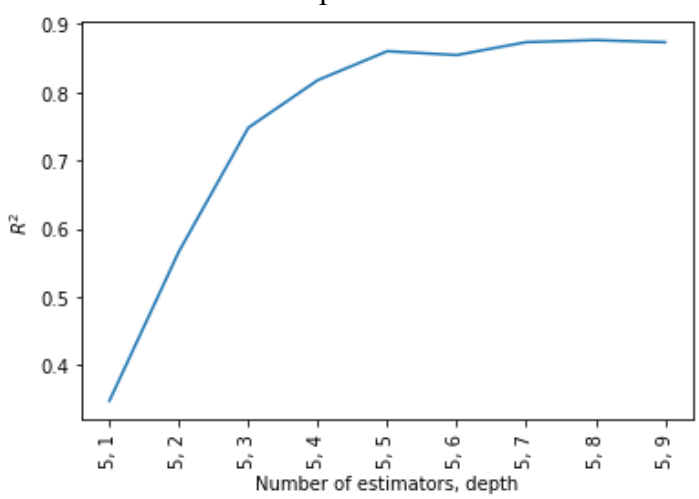

Figure 11. Hyperparameter Iteration Error Visualization

Below is an image of the model predictions using the best hyperparameters in orange against the actual values in blue from the dataset.

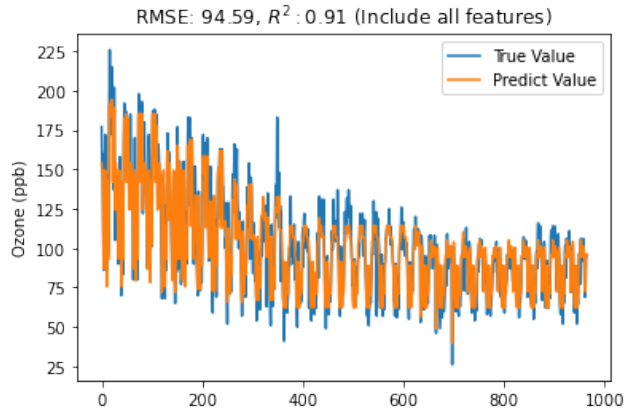

Figure 12. Actual (Blue) vs. Predicted (orange) Ozone Concentrations

The R^2 value using these hyperparameters was 0.9056 and the root mean squared error was 94.59. These values were found using the sci kit regression metrics tool kit. Also, we conduct a 10-fold cross validation to evaluate the uncertainty of the model and avoid overfitting. The R^2 value is 0.85 and RMSE value is 145.26. Our model clearly captures the trends of ozone concentration and peaks and troughs are aligned closely. Interestingly the model generally predicts below the absolute value of the actual concentration.

Conclusion

As we can see in table 2, Random Forest earns the best performance on our dataset under both RMSE and R^2 metric. Polynomial regression has higher R^2 than neural network while has poorer performance on RMSE. The 10-fold cross validation R^2 and RMSE of polynomial regression and random forest do not change dramatically compare to use whole dataset as input while neural network is more unstable compare these two models. For polynomial regression, it is intuitive to understand influence of each term. And sensitivity of each factor can be easily calculated by partial derivative. For random forest, it can capture some discrete relationships. The reason for the better performance of random forest among these three algorithms may be some potential threshold for specific conditions in ozone formation which may lead the relationships between ozone and features become discrete.

Table 2: Results of the three algorithms

	Polynomial Regression	Neural Network	Random Forest
RMSE	497.85	386.57	94.59
R^2	0.50	0.30	0.91
10-fold cross validation RMSE	503.60	470.23	145.26
10-fold cross validation R^2	0.49	0.17	0.85

There are several sources of uncertainty for our empirical model. Firstly, the emissions of VOC and NO_x are the main sources of the model. It is hard to have detailed and accurate emissions since some small emissions cannot be surveyed and measurements have relatively high uncertainty. Typically, the uncertainty of emissions is about 20%. Also, the lack of meteorology data for each site may cause uncertainty. In our project, limited meteorology data is available for ozone monitors. Using the nearest observed meteorological data for the ozone monitors could have problem since the meteorology condition can be varied from place to place.

For the future research, numerical model based on satellite data can be used to estimate weather conditions for each site and provide more creditable input data. Also, application of empirical model such as ozone isopleth and estimation of ozone exposure is also interested.

Reference

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